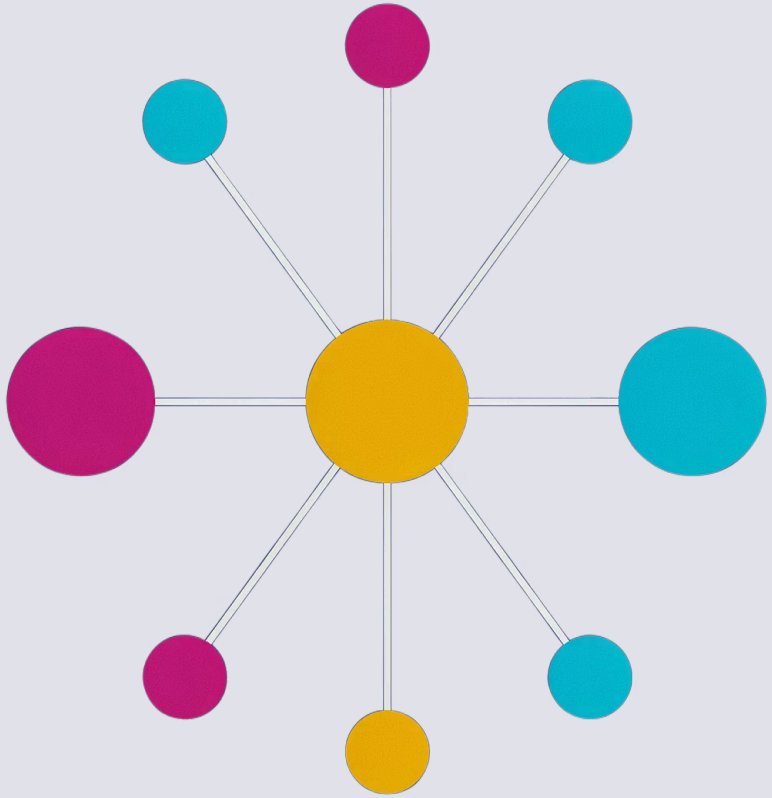
How a Transformer Understands Language

The one idea behind modern AI

Introduction to Generative AI





The One-Sentence Version

A Transformer figures out what a word means by **looking at its neighbours**, and it does this for **every word at once**, instead of one at a time.

Everything else (tokens, positional encoding, multi-head attention) is detail on top of this one idea.

Our Example Sentence

English:

"The bank can guarantee deposits."

French:

"La banque peut garantir les dépôts."

We will use this one sentence for the whole lesson.



The Ambiguity Problem

The word **"bank"** is ambiguous:

A river bank?

A financial bank?

The model has to work out which one from **context**, the same way a person would.

Three Steps to Attention

1. **The Problem:**
   how do *you* know what "bank" means?
2. **The Trick:**
   self-attention
3. **Reading vs Writing:**
   encoder and decoder

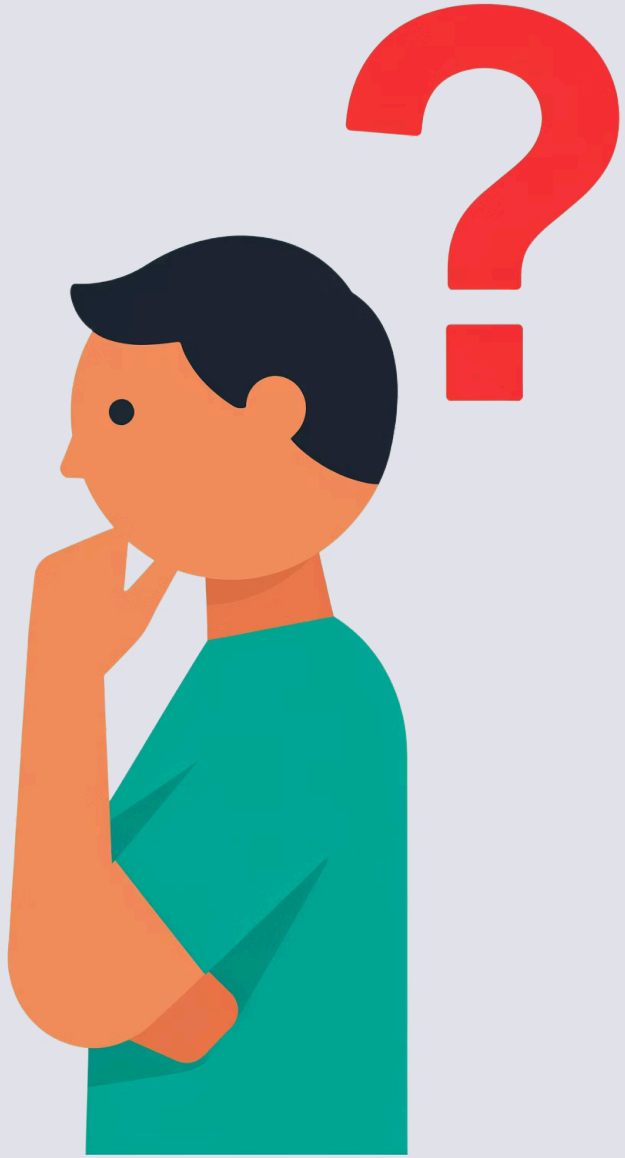
Step 1: The Problem

Look at the sentence again:

> "The bank can guarantee deposits."

Question for the class: does "bank" mean a river bank or a financial bank?

And the harder question: **how did you know?**

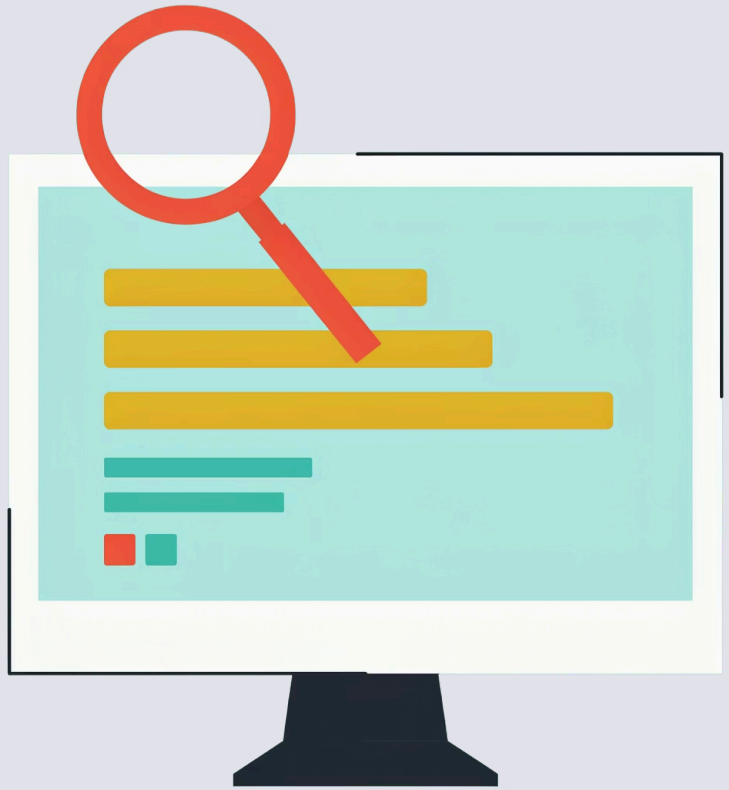


You Already Know Attention

Most people answer: *"Because of the word 'deposits'."*

You just described attention **without using the word**.

You used another word in the sentence to decide what "bank" means.



Step 2: The Trick (Self-Attention)

A computer does the same thing. For **every word**, it checks **every other word** and asks:

> "How much do you help me understand this word?"

Some words matter a lot. Some barely matter at all.

Scoring the Words

How relevant is each word to understanding **"bank"**?

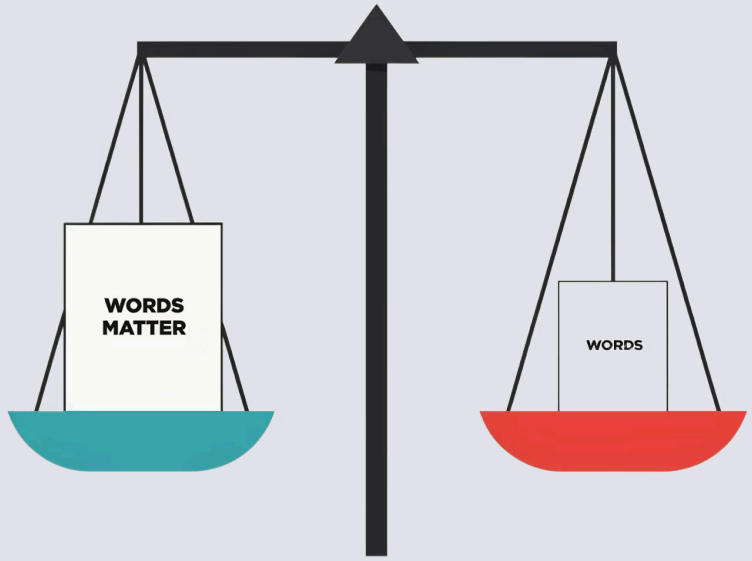
Word	Relevance to "bank"
The	Low
bank	Medium (itself)
can	Low
guarantee	**High**
deposits	**Very High**

Illustrative scores for teaching; real models learn these numbers from data.

The Conclusion

"deposits" scores **very high**, so the model concludes: this is a **financial** bank, not a river bank.

Scoring every word against every other word is called **Self-Attention**, the heart of the Transformer.



Step 3: Reading vs Writing

A librarian analogy:

The Encoder

reads and fully understands the English sentence first

The Decoder

then writes the French sentence **one word at a time**

Cross-Attention

while writing, the decoder glances back at the English whenever needed — that glancing back is called **Cross-Attention**



Translation Pipeline

Step	What happens
English sentence	The input you start with
Encoder	Reads and understands the full sentence
Decoder	Writes French one word at a time, using cross-attention to look back at the encoder
French translation	The final output

Common Misconception

Many students assume attention means paying attention to **nearby** words, like reading left to right, just faster.

That is wrong.



Why It Is Wrong



The model reads the **entire sentence at once**



That is exactly why it needs **position numbers** at all



If it read in order like a human, it would not need to be told the order

Worked Question

Why does "bank" pay so little attention to "can"?

"can" sits right next to "bank", only two words away.

If attention were about **closeness**, we would expect a thick line between them.



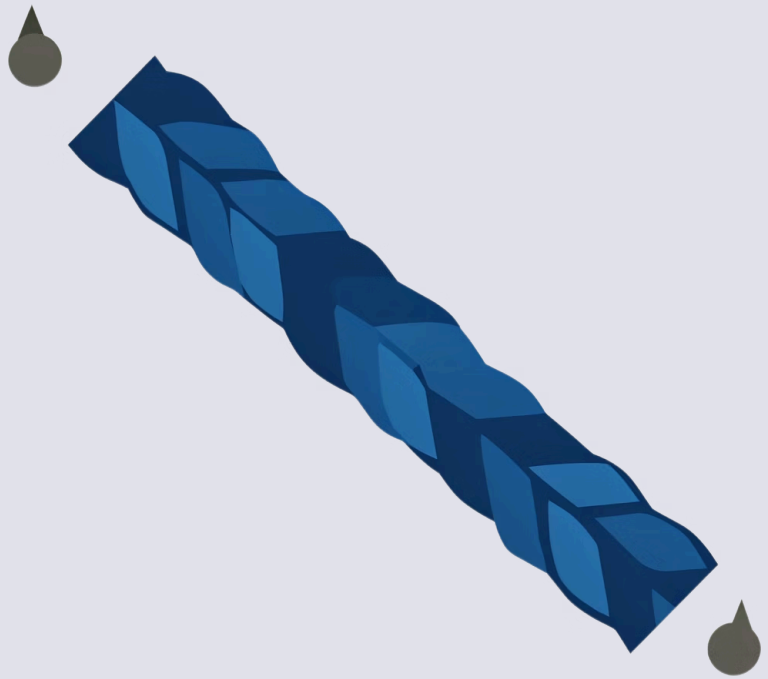
The Answer: "can"

- "can" is a modal verb doing **grammar work**, not **meaning work**
- It does not shift or narrow the meaning of "bank"
- It would appear identically in "The park bench can hold weight"
- So it carries almost no disambiguating information, and the model scores it **low**

The Answer: "deposits"

You **cannot swap** "deposits" out without changing what "bank" means.

That specificity is exactly what earns a **thick attention line**.



Takeaway

Attention is closer to:

> **"How much would knowing this word change my interpretation?"**

than to:

> "Which word is physically closest?"

A far-away word can get **high** attention. A next-door word can get **almost none**.

Quick Reference: Transformer Parts (1 of 2)

Part	What it does
Tokenization	Splits the text into small pieces (tokens)
Embeddings	Converts each piece into numbers a computer can use
Positional Encoding	Tells the model the order of the words
Encoder	Reads and understands the full input sentence

Quick Reference: Transformer Parts (2 of 2)

Part	What it does
Self-Attention	Figures out which words relate to which other words
Multi-Head Attention	Repeats the check several times, each looking for a different relationship
Decoder	Writes the output one word at a time
Cross-Attention	Looks back at the encoder to know what to translate next

Key Takeaways

1. A Transformer understands a word by looking at **the other words**, all at once
2. **Self-attention** scores how much each word helps explain another
3. Attention is about **meaning relevance**, not **distance**
4. The **encoder** reads; the **decoder** writes, using **cross-attention** to look back
5. **Positional encoding** exists because the model sees everything at once

Quick Check: Discuss in Pairs

1

In "The river bank was muddy", which word would get the highest attention from "bank"?

2

Why does the model need positional encoding if it reads everything at once?

3

What is the difference between self-attention and cross-attention?

4

Can a word far away from "bank" get high attention? Give an example.